
Bayesian Regression

§ 1.1

Bayesian Linear Regression

1.1.1 Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional feature vector associated with the i -th observation and $y_i \in \mathbb{R}$ denotes the corresponding scalar response.

Define design matrix $\mathbf{X}$, target vector $\mathbf{y}$, and coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

Under the linear regression model, each observation satisfies

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \epsilon_i,$$

where the noise terms are assumed to be independent and identically distributed as

$$\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N.$$

Define the noise vector

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix}.$$

It follows that

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$



where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix.

For the entire dataset, the linear regression model can therefore be written as

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}.$$

Consequently,

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) \sim \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N).$$

1.1.2 Inference

Bayesian method includes two basic steps: inference and prediction, where inference is to estimate posterior $\mathbf{w}$, while prediction is to predict y^* given $\mathbf{x}^*$.

Model Establishment

In Bayesian linear regression, the coefficient vector $\mathbf{w}$ is treated as a random variable and assigned a prior distribution. We assume that the prior distribution of $\mathbf{w}$ does not depend on $\mathbf{X}$, i.e., $w \perp X$, and thus $p(\mathbf{w} | \mathbf{X}) = p(\mathbf{w})$.

Given the training dataset, Bayesian formula is decomposed as:

$$p(\mathbf{w}|\mathcal{D}) = p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{X}, \mathbf{y})} = \frac{p(\mathbf{w}, \mathbf{y}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})} = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})} = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y}|\mathbf{X})},$$

where the fifth equality follows from

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}, \mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{X}, \mathbf{y})}{p(\mathbf{w}|\mathbf{X})p(\mathbf{X})} = \frac{p(\mathbf{w}, \mathbf{y}|\mathbf{X})}{p(\mathbf{w}|\mathbf{X})},$$

and the last equality follows from independency of prior coefficient and design matrix.

The denominator

$$p(\mathbf{y} | \mathbf{X}) = \int p(\mathbf{y} | \mathbf{X}, \mathbf{w})p(\mathbf{w})d\mathbf{w}$$

is the marginal likelihood. Since it does not depend on $\mathbf{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \propto p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}),$$

where $p(\mathbf{y}|\mathbf{X}, \mathbf{w})$ is a likelihood function and $p(\mathbf{w})$ is a prior function.

Suppose that a zero-mean Gaussian prior is imposed on the coefficient vector:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_p),$$

where $\boldsymbol{\Sigma}_p \in \mathbb{R}^{p \times p}$ denotes the prior covariance matrix.



Since both the likelihood and the prior are Gaussian, their conjugacy implies that the posterior distribution of $\mathbf{w}$ is also Gaussian:

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w)$$

Thus,

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \propto \mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_p).$$

The remaining task is therefore to derive the posterior mean $\boldsymbol{\mu}_w$ and posterior covariance $\boldsymbol{\Sigma}_w$.

Model Solution

To derive the posterior distribution, we first write the product of the likelihood and the prior explicitly:

$$\mathcal{N}(\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_p) = \frac{1}{(2\pi)^{N/2} \sigma^N} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})\right) \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}_p|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{w})^\top \boldsymbol{\Sigma}_p^{-1} \mathbf{w}\right).$$

Thus,

$$\begin{aligned} p(\mathbf{w}|\mathbf{X}, \mathbf{y}) &\propto \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}) - \frac{1}{2}(\mathbf{w})^\top \boldsymbol{\Sigma}_p^{-1} \mathbf{w}\right) \\ &= \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\mathbf{w} - \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w}) - \frac{1}{2}(\mathbf{w})^\top \boldsymbol{\Sigma}_p^{-1} \mathbf{w}\right) \\ &= \exp\left(-\frac{1}{2\sigma^2} \mathbf{y}^\top \mathbf{y} + \frac{1}{\sigma^2} \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} - \frac{1}{2} \mathbf{w}^\top \left(\frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}_p^{-1}\right) \mathbf{w}\right). \end{aligned}$$

Comparing this expression with the kernel of a multivariate Gaussian distribution,

$$\exp\left(-\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu}_w)^\top \boldsymbol{\Sigma}_w^{-1} (\mathbf{w} - \boldsymbol{\mu}_w)\right) = \exp\left(-\frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma}_w^{-1} \mathbf{w} + \mathbf{w}^\top \boldsymbol{\Sigma}_w^{-1} \boldsymbol{\mu}_w - \frac{1}{2} \boldsymbol{\mu}_w^\top \boldsymbol{\Sigma}_w^{-1} \boldsymbol{\mu}_w\right),$$

we obtain

$$\boldsymbol{\Sigma}_w^{-1} = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}_p^{-1} = \mathbf{A}.$$

Matching the linear terms gives

$$\mathbf{w}^\top \boldsymbol{\Sigma}_w^{-1} \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{w}^\top \mathbf{X}^\top \mathbf{y} \Leftrightarrow \mathbf{A} \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{y},$$

and therefore

$$\boldsymbol{\mu}_w = \frac{1}{\sigma^2} \mathbf{A}^{-1} \mathbf{X}^\top \mathbf{y}.$$

Hence, the posterior distribution has the closed-form solution

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) \sim \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w),$$



where

$$\Sigma_w^{-1} = \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1}, \quad \boldsymbol{\mu}_w = \frac{1}{\sigma^2} \Sigma_w \mathbf{X}^\top \mathbf{y}.$$

1.1.3 Prediction

After obtaining the posterior distribution of the coefficient vector, Bayesian linear regression can be used to predict the responses associated with new observations.

Let $\mathbf{X}^* \in \mathbb{R}^{N^* \times p}$ denote the design matrix for N^* new observations, and let $\mathbf{y}^* \in \mathbb{R}^{N^*}$ denote the corresponding target vector to be predicted.

Given the posterior distribution, the posterior predictive distribution is obtained by marginalizing over the uncertainty in $\mathbf{w}$:

$$p(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \int p(\mathbf{y}^* | \mathbf{X}^*, \mathbf{w}) p(\mathbf{w} | \mathcal{D}) d\mathbf{w}.$$

Thus, we have

$$p(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \mathcal{N}(\mathbf{X}^* \boldsymbol{\mu}_w, \mathbf{X}^* \Sigma_w (\mathbf{X}^*)^\top + \sigma^2 \mathbf{I}_{N^*}).$$

Therefore, the predictive mean and covariance are

$$\mathbb{E}[\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}] = \mathbf{X}^* \boldsymbol{\mu}_w,$$

and

$$\text{Cov}(\mathbf{y}^* | \mathbf{X}^*, \mathcal{D}) = \underbrace{\mathbf{X}^* \Sigma_w (\mathbf{X}^*)^\top}_{\text{parameter uncertainty}} + \underbrace{\sigma^2 \mathbf{I}_{N^*}}_{\text{observation noise}}.$$

§ 1.2

Bayesian Linear Regression

1.2.1 Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

consisting of N observations, where $\mathbf{x}_i \in \mathbb{R}^p$ denotes the p -dimensional feature vector associated with the i -th observation and $y_i \in \mathbb{B} = \{0, 1\}$ denotes the corresponding binary classification, like yes/no and success/failure.



Define design matrix $\mathbf{X}$, target vector $\mathbf{y}$, and coefficient vector $\mathbf{w}$ as

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N \times p}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^p.$$

The objective of logistic regression is to predict the probability $p(y_i = 1 | \mathbf{x}_i, \mathbf{w})$ rather than binary variable y_i .

Since the target response y_i is a binary variable, assume that

$$y_i | \mathbf{x}_i, \mathbf{w} \sim \text{Bernoulli}(p_i),$$

where

$$p_i = p(y_i = 1 | \mathbf{x}_i, \mathbf{w}), \quad 1 - p_i = p(y_i = 0 | \mathbf{x}_i, \mathbf{w}).$$

Recall that under the linear regression model, each observation satisfies

$$y_i = \mathbf{w}^\top \mathbf{x}_i + \epsilon_i.$$

However, the range of $\mathbf{w}^\top \mathbf{x}_i$ and p_i is different, where $\mathbf{w}^\top \mathbf{x}_i \in (-\infty, +\infty)$ and $p_i \in (0, 1)$. This required to find a suitable function that could map $\mathbf{w}^\top \mathbf{x}_i$ in $(-\infty, +\infty)$ to $(0, 1)$.

For logistic regression, the answer is logistic function (sigmoid function):

$$p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x}_i)} \in (0, 1).$$

Suppose that all observations are conditionally independent given $\mathbf{w}$, thus the likelihood of the whole dataset is

$$p(\mathbf{y} | \mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i} = \prod_{i=1}^N \sigma(\mathbf{w}^\top \mathbf{x}_i)^{y_i} (1 - \sigma(\mathbf{w}^\top \mathbf{x}_i))^{1-y_i}.$$

1.2.2 Inference

Bayesian method includes two basic steps: inference and prediction, where inference is to estimate posterior $\mathbf{w}$, while prediction is to predict y^* given $\mathbf{x}^*$.

Model Establishment

1.2.3 Prediction